

Population

Sample

Customer	Sales	Customer	Sales
C001	1200	C003	430
C002	850	C004	2200
C003	430	C005	1750
C004	2200	C006	980
C005	1750	C007	1500
C006	980	C008	760
C007	1500	C009	1340
C008	760	C010	890
C009	1340	C011	2100
C010	890	C012	640
C011	2100		1259
C012	640		
C013	1120		
C014	1580		
C015	720		
C016	1990		
C017	3050		
C018	470		
C019	880		
C020	1620		
C021	540		
C022	2450		
C023	910		
C024	1330		
C025	780		
C026	1560		
C027	990		
C028	1870		
C029	610		
C030	1420		
C031	520		
C032	1760		
C033	940		
C034	2680		
C035	830		
C036	1210		
C037	690		
C038	1540		
C039	1100		
C040	2300		
	1302.5		

Comparison

Is sample close to population?

Is sampling reliable?

Business decision

sample mean is 14 lower than the population mean.

14/1273 * 100 is 1.1 so it is close

yes but larger sample would be more reliable

Average customer spend is around ₹1250–₹1300

Column	Type	Type	Type	Type
CustomerID	Categorical	Discrete	Nominal	Nominal
Age	Numerical	Continuous	-	Ratio
Gender	Categorical	Discrete	Nominal	Nominal
City	Categorical	Discrete	Nominal	Nominal
SatisfactionLevel	Categorical	Discrete	Ordinal	Ordinal
MonthlySpend	Numerical	Continuous	-	Ratio

Month	Sales
M1	10000
M2	12000
M3	13000
M4	15000
M5	100000

Before

Mean 12500

Median 12500

Mode -

After

Mean 30000

Median 13000

Mode -

CustomerID	MonthlySpend
1	200
2	300
3	400
4	800

Range 600
Variance 51875
SD 227.7608395 227.7608

CustomerId	MonthlySpend
C1	2000
C2	2500
C3	3000
C4	3500
C5	4000
C6	4500
C7	5000
C8	5500
C9	6000
C10	15000

Mid Value = 4250

Q1 = 3000

q3 = 5500

IQR = 5500 - 3000 = 2500

Lower bound = -750

Upper bound = 9250

Outlier = 15000

$p = 25(10+1)/100 = 2.75$

So 2.75th position

CustomerID	Phone	Accessories
1	1	0
2	1	1
3	0	1
4	1	1
5	0	0
6	1	0
7	0	1
8	1	1
9	0	0
10	1	1

P(Phone)

0.6

P(Accessories)

0.6

P(Phone AND Accessories)

0.4

P(Phone OR Accessories)

0.8

P(Phone | Accessories)

0.66666667

CustomerID	BoughtPhone	BoughtAccessories	HighIncome
1	1	1	1
2	1	0	0
3	0	1	1
4	1	1	0
5	0	0	0
6	1	1	1
7	0	1	0
8	1	0	1
9	0	0	0
10	1	1	1

P(Phone)

0.6

P(Accessories)

0.6

P(Phone AND Accessories)

0.4

P(Phone | Accessories)

0.66666667

P(Accessories | Phone)

0.66666667

0.66666667

Identify High-Probability Segments

4

Recommend targeting strategy

0.4

CustomerID	Segment	Spend
1	A	500
2	B	600
3	A	550
4	C	700
5	B	650
6	A	520
7	C	720
8	B	680
9	A	510
10	C	750

Population Mean

618

Random Sample Mean (pick any 5 rows)

600

Stratified (Segment A only)

520

Systematic (every 2nd row)

586

Date	SalesBefore	SalesAfter
1	100	120
2	110	130
3	105	125
4	115	140
5	120	150
6	130	160
7	125	155
8	135	165
9	140	170
10	150	180

Compute average sales before and after

Average Before

123

Average After

149.5

Perform hypothesis test (1 → one-tailed test

1.3519E-08 <----- 1 → paired test

Calculate p-value

1.3519E-08

Decide whether campaign improved sales

Decision

p-value < 0.05

Reject H_0

Campaign significantly improved sales

CustomerID	RevenueBefore	RevenueAfter
1	200	240
2	220	260
3	210	250
4	230	280
5	240	300
6	250	310
7	245	305
8	260	320
9	270	330
10	280	350

Define Hypotheses

H₀ (Null Hypothesis):

There is no change in revenue (Before = After)

H₁ (Alternative Hypothesis):

Revenue increased after campaign (After > Before)

Calculate Average Revenue

Average Before

240.5

Average After

294.5

Perform Paired t-test

3.42436E-08

Compute p-value

3.42436E-08

Interpret Result

p-value < 0.05

Reject H₀

Campaign significantly increased revenue

CustomerID	Gender	ProductCategory
1	Male	Electronics
2	Female	Clothing
3	Male	Electronics
4	Female	Clothing
5	Male	Groceries
6	Female	Electronics
7	Male	Clothing
8	Female	Groceries
9	Male	Electronics
10	Female	Clothing
11	Male	Groceries
12	Female	Electronics
13	Male	Clothing
14	Female	Groceries
15	Male	Electronics

Gender	Electronics	Clothing	Groceries	Total
Male	4	2	2	8
Female	2	3	2	7
Total	6	5	4	15

Calculate expected frequencies

3.2

Gender	Electronics	Clothing	Groceries
Male	3.2	2.67	2.13
Female	2.8	2.33	1.87

Compute χ^2 Statistic

0.81

Compute p-value

0.666976811

Final Decision

p-value = 0.66 > 0.05

Fail to reject H_0

No significant relationship

CustomerID	AgeGroup	PurchaseValue
1	Young	200
2	Young	220
3	Young	210
4	Middle	300
5	Middle	320
6	Middle	310
7	Senior	400
8	Senior	420
9	Senior	410

Mean Purchase Value for Each Group

Young

210

Middle

310

Senior

410

Overall Mean

310

Perform One-Way ANOVA

Step 1: Between-Group Variance (SSB)

Young

$$= 3(210-310)^2 = 3(-100)^2 = 3 \times 10000 = \mathbf{30000}$$

Middle

$$= 3(310-310)^2 = 0$$

Senior

$$= 3(410-310)^2 = 3 \times 10000 = \mathbf{30000}$$

$$\mathbf{SSB = 60000}$$

Step 2: Within-Group Variance (SSW)

Young

$$(200-210)^2 + (220-210)^2 + (210-210)^2$$

$$100 + 100 + 0 = 200$$

Middle

$$100 + 100 + 0 = 200$$

Senior

$$100 + 100 + 0 = 200$$

$$\mathbf{SSW = 600}$$

F-Statistic

$$F = 30000 / 100 = \mathbf{300}$$

p-value

9.7059E-07

Final Decision

p-value < 0.05

Reject H_0

Age groups differ significantly

Sales	AdvertisingSpend	ProductPrice	CustomerVisits
200	50	20	300
250	60	22	320
300	70	25	350
350	80	27	370
400	90	30	400
450	100	32	420
500	110	35	450
550	120	37	470
600	130	40	500
650	140	42	520

Compute Correlation Matrix

Sales vs Advertising

1

Sales vs Price

0.999406

Sales vs Visits

0.999406

AdvertisingSpend	Sales
10	100
20	150
30	200
40	250
50	300
60	350
70	400
80	450
90	500
100	550

Predict Future Sales

For Advertising = 120

650

CustomerID	Age	Income	SpendingScore
1	25	50000	60
2	30	60000	70
3	35	55000	65

Represent Each Customer as Vector

Customer 1 $\rightarrow A = [25, 50000, 60]$

Customer 2 $\rightarrow B = [30, 60000, 70]$

Customer 3 $\rightarrow C = [35, 55000, 65]$

Compute Vector Addition ($A + B$)

Calculation:

$[25+30, 50000+60000, 60+70]$

$[55, 110000, 130]$

Compute Dot Product ($A \cdot B$)

3000004950

Compute L2 Norm of A

50000.0422

Distance Between A and B

10000.0062

Compute L2 Norm of B

60000.0483

Cosine Similarity

1

Age	Income	Spending
22	30	150
25	35	200
30	50	250

Represent Dataset as Matrix

Matrix 22 30 150
 25 35 200
 30 50 250

Identify Matrix Shape

Shape = (3 × 3)

Multiply Matrix by Scalar (2×)

2A = 44 60 300
 50 70 400
 60 100 500

Add Two Matrices

A+A = 44 60 300
 50 70 400
 60 100 500

Interpret Rows and Columns

Rows → Each row represents one customer

Row 1 → Customer 1 (Age 22, Income 30, Spending 150)

Columns → Each column represents a feature

Column 1 → Age

Column 2 → Income

Column 3 → Spending

Age	Income
25	40
30	50
35	60

Represent Dataset as Matrix

A = 25 40
 30 50
 35 60

Create Transformation Matrix

T = 1 0
 0 2

Perform Matrix Multiplication (A × T)

A * T = 25 80
 30 100
 35 120

Compute Transpose of Dataset

A^T = 25 30 35
 40 50 60

$$3x+2y=16$$

$$x+4y=14$$

Convert to Matrix Form ($Ax = b$)

$$\begin{bmatrix} 3 & 2 \\ 1 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 16 \\ 14 \end{bmatrix}$$

$$A = \begin{bmatrix} 3 & 2 \\ 1 & 4 \end{bmatrix}$$

$$x = \begin{bmatrix} x \\ y \end{bmatrix}$$

$$B = \begin{bmatrix} 16 \\ 14 \end{bmatrix}$$

Compute Determinant of A

$$|A| = 10$$

Find Matrix Inverse

$$A^{-1} = \begin{bmatrix} 0.4 & -0.2 \\ -0.1 & 0.3 \end{bmatrix}$$

Compute Solution $x=A^{-1}b$

$$X = \begin{bmatrix} 3.6 \\ 2.6 \end{bmatrix}$$